CAUSAL-EFFECT MEMORY FOR RAW VISUAL EVENTS IN FINITE-WINDOW WORLD MODELS

Anonymous authors
Paper under double-blind review

ABSTRACT

Finite-window world models cannot directly use observations after they leave the context window. We introduce Causal-Effect Memory (CEM), an external controller that proposes events from prediction surprise, verifies them using future-latent loss, stores verified event versions, and retrieves a bounded set for later prediction. The controller receives no event labels or event times. We evaluate CEM on unmodified OGBench renderings using frozen DINOv2 features and action-conditioned latent predictors, with separate controlled studies on released DINO-WM and LeWorldModel hosts. Across diverse OGBench environments, memory consistently improves prediction over no-memory, reset, shuffled, and random-memory controls, and its benefit grows with rollout horizon. However, a recent-only extension is slightly stronger on average, and learned causal-effect rankings remain weak. The released DINO-WM study provides stronger deletion evidence, while the LeWorldModel study shows that a readable memory can still be incompatible with a frozen predictor. These results support external event memory as a useful but limited extension to finite-window world models and identify event ranking and memory-interface compatibility as the main unresolved problems.

1 INTRODUCTION

World models often use a fixed observation window. An observation can affect the future after it leaves this window, but the model can no longer attend to it directly. Extending the window increases cost and does not decide which past observations are useful. A practical external memory must instead answer three questions: when to write, what to retain, and what to retrieve for a particular prediction.

We study this problem without supplying event times or event labels. Causal-Effect Memory (CEM) uses the prediction loss of a fixed world model as its training and evaluation signal. Prediction surprise proposes a visual event. A delayed test measures whether retaining the event reduces action-conditioned future-latent loss. A learned score approximates this measurement, and periodic group deletion updates the score. At inference, a bounded router ranks verified events using query content, event content, age, and predicted need.

Our main experiment uses original OGBench renderings. The model receives only frames, actions, and timestamps. Frozen DINOv2 patch tokens provide visual features, and an action-conditioned latent predictor is trained on each environment and then frozen. These predictors are *DINO-feature action-conditioned world models*; they follow the feature-space design of DINO-WM (Zhou et al., 2025), but they are not released DINO-WM checkpoints. The study covers 10 environments from navigation, manipulation, puzzle, and scene families.

The raw results are positive but limited. CEM lowers four-step prediction error relative to no-memory, reset, shuffled-episode, and matched-random controls in every environment, and the improvement grows with rollout horizon. A recent-only extension is nevertheless slightly better on average. Moreover, the learned causal-effect ranking has low held-out correlation with measured deletion effects. Thus past state is useful, but the current method does not reliably identify the most useful event.

Two controlled studies clarify this result. First, the released frozen DINO-WM Wall host shows a large gap between selected-event and random-event deletion under a separate encoded event-versioning protocol. Second, a frozen LeWorldModel (LeWM) (Maes et al., 2026) can retain a six-way binding in memory while failing to use it without harming its original prediction loss. We call *host readiness* the ability of a pretrained world model to use an external memory representation. After this definition, we use the more direct term *memory interface compatibility*.

Contributions.

- We define WRITE, KEEP, and RECALL with one causal importance score: the change in a fixed host's future-latent loss when an event group is removed.
- We implement a label-free raw-image protocol with frozen DINOv2 patch tokens, action-conditioned host surprise, delayed group verification, event versioning, hysteresis, and bounded retrieval.
- We report trajectory-disjoint results for 18 cells, 10 environments, and four families, including reset, shuffled, random, recent-only, budget, delay, rollout, deletion, and action-sequence controls.
- We separate DINO-feature breadth from released DINO-WM validation and show that storage quality and memory interface compatibility are distinct.

2 Related Work

Predictive world models. Joint-embedding predictive architectures learn by predicting latent targets rather than pixels (Assran et al., 2023). DINO-WM predicts frozen DINO features and uses them for planning (Zhou et al., 2025); LeWM trains an end-to-end latent world model (Maes et al., 2026). Causal-JEPA studies object-level latent masking (Anonymous, 2026). These methods learn predictive representations inside a configured context. We study an external state that persists after an observation leaves that context.

Learned memory. Recurrent networks, state-space models, and slot memories provide different ways to compress history (Cho et al., 2014; Hochreiter and Schmidhuber, 1997; Gu and Dao, 2023; Jiang et al., 2024; Zhang et al., 2024). SlotMemory uses object-centered key–value storage (Anonymous, 2026). Titans writes to neural memory using surprise (Behrouz et al., 2025), and test-time training adapts computation online (Sun et al., 2024). In CEM, surprise only proposes an event. Retention is trained against the effect of removing the event from a fixed external host.

Memory evaluation. RULER and LongMemEval show that nominal context length can overstate usable memory (Hsieh et al., 2024; Wu et al., 2025). "On Memory" argues that memory should be evaluated through downstream use (Laird and Clark, 2025). We use a *controlled memory evaluation*: full, reset, recent-only, shuffled, and random memories share the same host, data, and prediction target. Probe accuracy is secondary to host prediction loss and deletion tests.

3 Problem Definition

Let a fixed host encode frame x_t as z_t and predict future latents from a legal context of K observations and actions. A memory store $\mathcal{M}_t$ contains event groups

$$m_i = (k_i, v_i, t_i, \widehat{\mathrm{CE}}_i, \ell_i),$$

where k_i and v_i are a semantic key and value, t_i is the observed timestamp, $\widehat{\mathrm{CE}}_i$ is a predicted causal importance score, and ℓ_i records whether the event is provisional, active, superseded, rejected, or evicted.

For future evaluation set $\mathcal{H}_t$, define the measured group effect

$$\mathrm{CE}(G) = \mathbb{E}_{u \in \mathcal{H}_t} [\mathcal{L}_{\mathrm{future}}(\mathcal{M}_t \setminus G) - \mathcal{L}_{\mathrm{future}}(\mathcal{M}_t)] \,. \quad (1)$$

The loss is the host's action-conditioned latent prediction error. We delete adjacent frames as a group because one visual transition can span several frames. Equation 1 measures an intervention on the memory input to a fixed computation. It is not an estimate of a causal effect in the external environment.

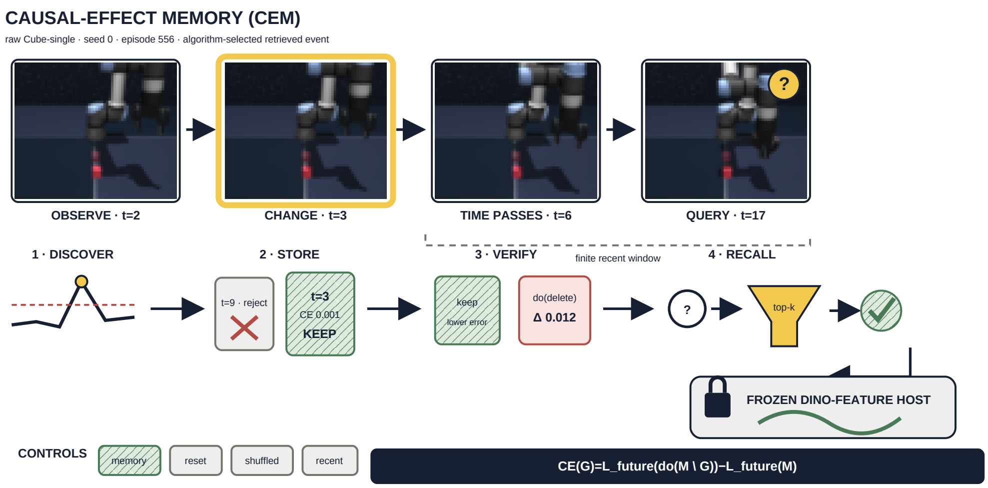

Figure 1: CEM on an unmodified OGBench trajectory. Four raw Cube-single frames are selected by a fixed pixel-change plus DINO-change score, restricted to retrieved test events. CEM discovers the t=3 change, rejects a later low-value event, verifies the stored group by deletion, and recalls it at t=17. No event time or label is supplied. Exact scores and hashes are in `generated_results/architecture_example_receipt.json`.

4 CAUSAL-EFFECT MEMORY

4.1 WRITE: SURPRISE AND SEMANTIC CHANGE

The fixed host predicts $\hat{z}_t$ from the legal context and action. Its one-step surprise is

$$s_t = \|\hat{z}_t - z_t\|_2^2.$$

Separately, the method measures temporal change in frozen DINOv2 patch-token features. Train-set quantiles set both thresholds. Adjacent threshold crossings form a provisional event group, and the highest combined crossing defines its timestamp. This procedure has no manually chosen interval, color target, or saliency function.

4.2 KEEP: DELAYED GROUP VERIFICATION

A small estimator predicts the future-loss reduction from retaining a provisional group. Training targets are computed periodically by comparing the host with a singleton event group against the same host with no event. The target is normalized by no-memory loss. Smooth-L1 regression provides scale calibration, and a pairwise ranking loss trains ordering within each query. A group is promoted only after the verification delay and only when its predicted score exceeds a validation-selected threshold.

4.3 EVENT VERSIONING

Event versioning means that semantically similar events share a key but keep separate timestamps and scores. Train-only clustering of DINO event vectors assigns keys. A new candidate does not remove an active event when it is proposed. The old event remains available until the new candidate passes verification by more than a hysteresis margin. If verification fails, the store falls back to the old version. Capacity eviction removes the lowest scored active event.

4.4 RECALL: BOUNDED ROUTING

For query q_t, the router scores active events by

$$r_i = q_t^\top k_i + \phi(t - t_i) + \nu(q_t, \text{context}) + \lambda \widehat{\text{CE}}_i, \tag{2}$$

where ϕ is a learned age function and ν predicts whether the current query needs memory. The router retrieves at most k events. Verification uses the measured singleton group loss after the

Algorithm 1: Online CEM with delayed verification

Input: Frozen host h, bounded store $\mathcal{M}$, stream (x_t, a_t)

$z_t \leftarrow h_{\text{enc}}(x_t)$; compute one-step surprise and DINO change;
if *train-calibrated thresholds cross* **then**
    append adjacent frames as a provisional group
end
foreach *group whose verification delay expires* **do**
    predict $\widehat{\text{CE}}(G)$; promote, reject, or supersede by hysteresis;
end
evict the lowest-value active group if $|\mathcal{M}|$ exceeds budget;
$C_t \leftarrow \text{topk}_{G \in \mathcal{M}}\, r(G, q_t)$;
return frozen-host prediction conditioned on C_t;
Audit only after the target arrives: measure hard-delete $\text{CE}(G)$;

Table 1: **Signal-to-claim contract.** Each stage supports a narrower claim than the next; only hard deletion through the frozen host is treated as causal evidence.

signal	controller role	licensed interpretation
surprise + DINO change	DISCOVER	candidate, not importance
$\widehat{\text{CE}}$	STORE / replace	amortized predicted value
query/content/age/need	RECALL	bounded relevance ranking
hard group deletion	VERIFY / audit	effect on fixed-host prediction

future evaluation horizon. We report this bounded retrieve-then-verify audit separately. The primary prediction uses the routed event before this outcome is known, so its test target never selects its own memory.

Training/evaluation separation. Thresholds, key clustering, the effect estimator, and router are fit on train trajectories and selected on validation trajectories. Test futures never select the primary memory. The retrieve-then-verify result is a separate post-hoc audit because it observes the realized future target.

Bounded cost. Surprise is a by-product of the host forward pass. Online KEEP and eviction use the amortized score; true group deletion is sampled during calibration and audit. The store, top-k read, and version budget are fixed, so persistent state does not enlarge the host's legal attention window.

5 Experimental Protocol

5.1 Raw OGBench breadth

We use original 64×64 render caches with 768 trajectories and 22 frames per environment. The environments are PointMaze-large, PointMaze-teleport, PointMaze-giant, Cube-single, Cube-double, Cube-triple, Puzzle-3x3, Scene, AntMaze-large, and HumanoidMaze-large. PointMaze-large, PointMaze-teleport, Cube-single, and Cube-triple use three optimization seeds; the other environments use one breadth seed.

A fixed trajectory permutation creates 538 train, 115 validation, and 115 test trajectories. A frozen DINOv2 ViT-S/14 produces a spatial pyramid of patch tokens. A 96-dimensional PCA projection is fit on train trajectories only. For each environment and seed, a finite-window action-conditioned predictor is trained on train trajectories, selected by validation loss, and frozen before CEM training. All 18 hosts beat latent persistence on test.

The model reads only 'frames', 'actions', and array timestamps. Other cache keys are ignored. The validation split selects write thresholds, verification delay, hysteresis, budget, and retrieval count. The test split reports future-latent MSE by horizon, all memory controls, causal deletion, score calibration, write and retrieval rates, budget and delay curves, and post-hoc action-sequence identification.

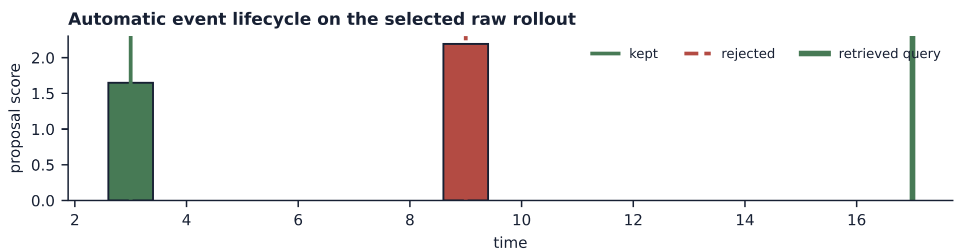

Figure 2: One automatic decision trace. On the same raw Cube rollout used in Fig. 1, the t=3 event is verified and later retrieved at t=17. A larger proposal at t=9 is rejected by delayed verification. Proposal strength therefore does not determine retention.

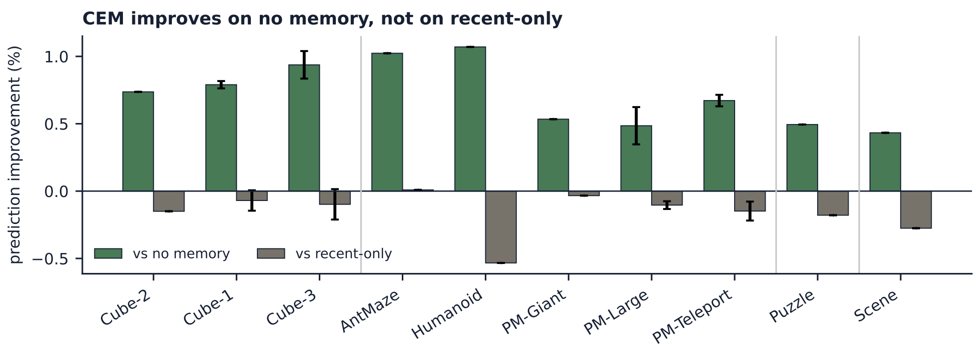

Figure 3: Does CEM improve future prediction? Per-environment four-step prediction improvement, normalized to each control. CEM improves on no memory in all ten environments, but recent-only is better in 15/18 optimization cells. Error bars show optimization-seed variation where available; separators denote manipulation, navigation, puzzle, and scene.

5.2 Separate controlled protocols

The released DINO-WM Wall protocol freezes the published predictor, action/proprioception encoders, and DINOv2 features. It evaluates event versioning by relocating cached encoded events in time. This is a controlled representation-level protocol, not raw OGBench breadth or planning.

The LeWM PushT protocol freezes the released encoder and predictor and tests a six-way same-base visual binding. It measures the path from readable memory to the predictor output. Semantic class labels are used only by post-hoc readouts. These synthetic tasks diagnose overwrite, binding, and interface failures; they are not the main breadth evidence.

5.3 Decision anatomy

The trace separates DISCOVER from KEEP. The later event is more surprising, but its predicted future value is below the active version's hysteresis threshold. CEM retains the older event and retrieves it without a supplied event boundary. The deterministic selection score and raw frame hashes are in `generated_results/architecture_example_receipt.json`.

6 Results

6.1 Raw OGBench breadth

CEM lowers prediction error relative to no memory in every environment. The mean reduction is 0.719% and grows from 0.227% at rollout step one to 1.086% at step four. Reset, shuffled, and matched-random memories do not explain the gain. The stronger recent-only control does: CEM is slightly worse on average and wins only 3/18 cells. The primary result is therefore useful persistent state, not superior event selection. Raw MSE values and horizon curves are reported in Appendix E.

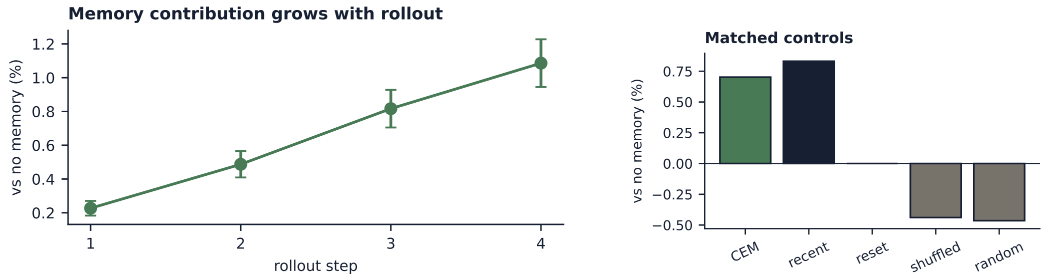

Figure 4: When does memory help? Left: normalized improvement grows over the four-step rollout. Right: matched controls confirm that an arbitrary residual does not help, while recent-only remains the strongest alternative. Intervals treat the 18 optimization cells as observations.

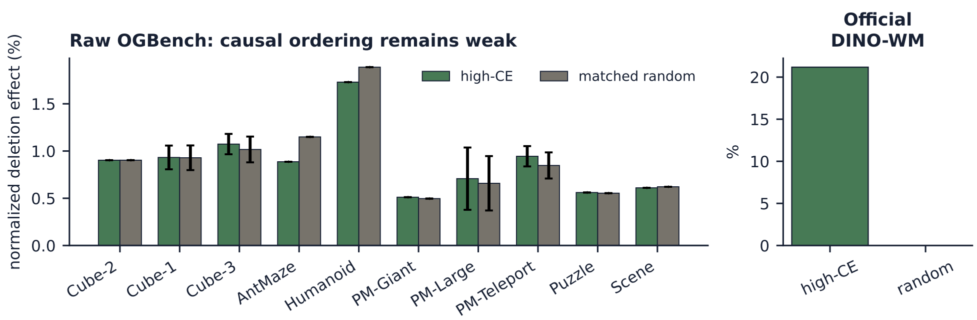

Figure 5: Does the learned score identify causal events? Deletion effects are normalized by each host's no-memory prediction error. On raw OGBench, high-CE and matched-random groups have similar effects and vary by environment. The released DINO-WM protocol yields a large selected-versus-random contrast, but is a separate encoded-event study.

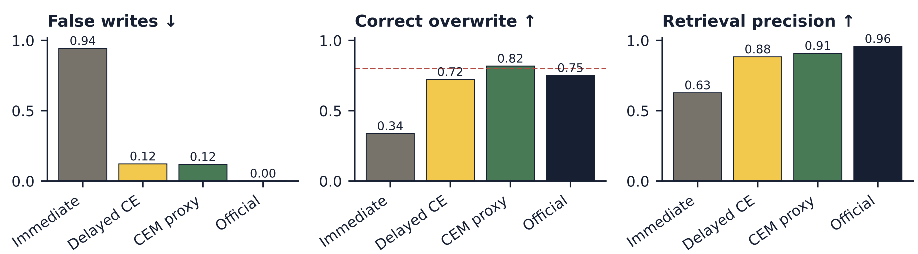

Figure 6: Which controller decisions matter? Immediate surprise, delayed causal-effect verification, the complete proxy controller, and released DINO-WM are compared on false writes, correct overwrite, and retrieval precision. Proxy bars average Cube and PointMaze; the official bar is a separate Wall protocol.

6.2 Causal ranking is not yet reliable

High-ranked deletion exceeds matched random deletion in 12/18 cells, but the aggregate gap includes zero and held-out rank correlation is near zero. CEM therefore discovers useful state more reliably than it orders individual events. Calibration and action-sequence diagnostics are reported in Appendix F; neither supports a planning claim.

6.3 Delayed verification controls the event store

Immediate surprise writes distractors. Delayed group verification sharply reduces false writes and improves overwrite and retrieval. Event versions then protect an active event while a replacement is tested. These task-level rates motivate the full controller without relying on raw prediction-loss units.

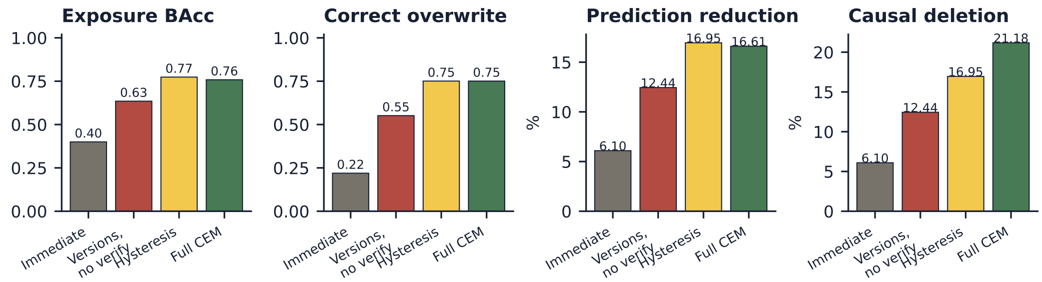

Figure 7: Released frozen DINO-WM Wall factorial (three seeds). The panels report task exposure, correct overwrite, normalized prediction-error reduction, and selected-event deletion relative to the reset baseline. This encoded controlled protocol is separate from raw OGBench breadth.

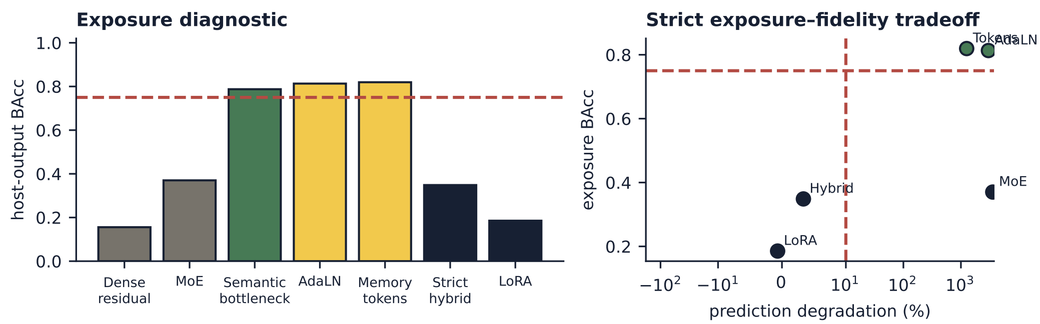

Figure 8: Frozen LeWM interface study (six-way PushT binding, age 15, three seeds). Left: task information available at the host output. Right: relative prediction degradation for interfaces evaluated with the same strict target. Non-comparable legacy objectives appear only in the left diagnostic.

6.4 Released DINO-WM gives stronger controlled deletion evidence

On the released DINO-WM host, CEM exposes the event above chance, improves prediction by 16.6%, and produces a 21.2% normalized selected-event deletion effect while random deletion is zero. This is clear host-use evidence, but the protocol relocates cached encoded events and does not establish raw breadth or planning.

6.5 LeWM reveals memory interface incompatibility

The memory state retains the six-way binding, but dense residual injection loses it before the frozen predictor. Memory tokens and AdaLN expose the binding, but severely degrade prediction. Strict hybrid and LoRA preserve prediction but remain near chance. No tested point combines high exposure with less than 10% degradation. Storage quality and host-interface compatibility are therefore separate.

7 Interpretation

Prediction and causal ordering are distinct. An imperfect event set can lower average prediction error even when its within-query ranking is weak. CEM is consistent on the former endpoint and mixed on the latter, which is why the raw breadth and deletion figures answer separate questions.

Recency is the strongest alternative. Reset, shuffled, and random controls rule out a generic residual benefit. Recent-only retains valid short-range trajectory state and remains stronger. A decisive next benchmark must break local continuity with longer, trajectory-disjoint gaps while preserving the same memory and retrieval budget.

Breadth is directional, not equally precise. Four representative environments have three optimization seeds; six have one. The ten-environment result establishes a shared direction over no memory,

Table 2: Evidence is reported at four distinct levels. A stronger claim is not inferred from a weaker endpoint.

claim	status	evidence
raw state improves on no memory	supported	10/10 environments; matched controls
CEM beats local recency	unsupported	3/18 cell wins
learned CE orders raw events	unresolved	deletion-gap interval includes zero
released host uses selected memory	supported	fail-closed audit + deletion
frozen LeWM is memory-compatible	unsupported	exposure–fidelity trade-off

not equal confidence across families. Full per-environment losses, uncertainty, telemetry, and audit receipts remain in the appendix.

8 Claim Hierarchy

Why the hierarchy matters. Prediction improvement establishes that retrieved history can be useful. It does not establish that the written event is uniquely necessary, that the learned score is calibrated, or that an arbitrary pretrained host can consume the memory. The raw, official-host, and LeWM protocols isolate these questions rather than averaging them into one headline metric.

What would change the conclusion. The raw selection claim would require CEM to outperform a capacity-matched recent-only state across longer gaps and to show a positive high-versus-random deletion interval. The host-compatibility claim would require a LeWM interface above the exposure gate while remaining below the registered prediction degradation bound. Neither condition is met here.

Why the official result remains separate. Released DINO-WM supplies the clearest selected-event deletion contrast, but it uses relocated cached latent events. It validates the controller's causal-use mechanism under a fixed released host; it cannot substitute for raw-event breadth, automatic semantic discovery, or executed planning.

9 Discussion and Limitations

What the raw result establishes. Frozen DINO features, host surprise, and a small event store can improve action-conditioned prediction without supplied event labels or times. The increasing horizon gain and the shuffled/random controls support a genuine use of past visual state. The result covers unmodified simulated environment renderings and does not test real-world camera data.

What it does not establish. Recent-only state is stronger on average, CE-score calibration is low, and the deletion-ordering interval includes zero. Trajectories contain only 22 frames and four rollout steps are evaluated. Six environments have one breadth seed. The PCA projection and memory adapter are environment-specific. These limits rule out claims of unbounded recall, general causal discovery, or transfer.

Host dependence. Equation 1 can identify only information represented and used by the fixed host. A blind host produces no useful score. Conversely, a memory can retain information that the host interface removes, as in LeWM. Future work should jointly train a memory-compatible host interface while retaining controlled reset, shuffled, random, and recent-only comparisons.

Computation. Periodic group deletion requires extra host evaluations, and bounded retrieve-then-verify scales with the number of routed candidates. The current implementation limits this cost with amortized scoring and top-k retrieval, but does not report a standardized cross-host latency benchmark.

10 Conclusion

CEM extends a finite-window world model with label-free visual event memory whose decisions are trained against future-latent prediction loss. On raw OGBench renderings, it consistently improves over no memory and corruption controls, with larger gains at longer rollout horizons. It does not yet beat a recent-only extension or reliably rank event effects. Released DINO-WM and LeWM diagnostics show why both event selection and memory interface compatibility must be evaluated. These results provide a reproducible baseline for raw visual event memory and a clear set of remaining failures.

References

Mahmoud Assran, Quentin Duval, Ishan Misra, Piotr Bojanowski, Pascal Vincent, Michael Rabbat, Yann LeCun, and Nicolas Ballas. Self-supervised learning from images with a joint-embedding predictive architecture. In *CVPR*, 2023.

Gaoyue Zhou, Hengkai Pan, Yann LeCun, and Lerrel Pinto. DINO-WM: World models on pre-trained visual features enable zero-shot planning. In *ICML*, 2025.

Lucas Maes, Quentin Le Lidec, Damien Scieur, Yann LeCun, and Randall Balestriero. LeWorldModel: Stable end-to-end joint-embedding predictive architecture from pixels. *arXiv:2603.19312*, 2026.

Anonymous. Causal-JEPA: Learning world models through object-level latent masking. *arXiv:2602.11389*, 2026.

Kyunghyun Cho, Bart van Merrienboer, Caglar Gulcehre, Dzmitry Bahdanau, Fethi Bougares, Holger Schwenk, and Yoshua Bengio. Learning phrase representations using RNN encoder-decoder for statistical machine translation. In *EMNLP*, 2014.

Sepp Hochreiter and Jürgen Schmidhuber. Long short-term memory. *Neural Computation*, 9(8):1735–1780, 1997.

Albert Gu and Tri Dao. Mamba: Linear-time sequence modeling with selective state spaces. *arXiv:2312.00752*, 2023.

Jindong Jiang, Fei Deng, Gautam Singh, Minseung Lee, and Sungjin Ahn. Slot state space models. In *NeurIPS*, 2024.

Yu Zhang et al. Gated slot attention for efficient linear-time sequence modeling. In *NeurIPS*, 2024.

Anonymous. SlotMemory: Object-centric KV memory for streaming long-video generation. *arXiv:2605.31033*, 2026.

Ali Behrouz, Peilin Zhong, and Vahab Mirrokni. Titans: Learning to memorize at test time. *arXiv:2501.00663*, 2025.

Yu Sun et al. Learning to (learn at test time): RNNs with expressive hidden states. *arXiv:2407.04620*, 2024.

Cheng-Ping Hsieh, Simeng Sun, Samuel Kriman, Shantanu Acharya, Dima Rekesh, Fei Jia, Yang Zhang, and Boris Ginsburg. RULER: What's the real context size of your long-context language models? In *COLM*, 2024.

Di Wu, Hongwei Wang, Wenhao Yu, Yuwei Zhang, Kai-Wei Chang, and Dong Yu. LongMemEval: Benchmarking chat assistants on long-term interactive memory. In *ICLR*, 2025.

Eli J. Laird and Corey Clark. On memory: A comparison of memory mechanisms in world models. *arXiv:2512.06983*, 2025.

A Artifact Map and Generation Contract

Every paper value is read from machine-readable artifacts by `scripts/plot_paper_d.py`. The script writes white-background figures, table fragments, the raw-frame selection receipt, and paper snapshots. Generated paper files are self-contained, so compilation does not require the gitignored output tree.

Table 3: Protocol and artifact map. "Official" denotes a released host checkpoint. The DINO-feature breadth host is trained locally and is not official DINO-WM.

protocol	host/tasks	source	paper products
Raw breadth	DINO-feature host, 10 OGBench environments	`outputs/cem_raw_ogbench/report.json`; `outputs/cem_raw_ogbench/cells/**/result.json`	Figs. 1–5
Official validation	released DINO-WM, Wall	`outputs/cem_event_versioning_dinowm_official_v1/report.json`	Fig. 7
Memory interface	released LeWM, PushT	`outputs/cem_lewm_\{semantic_adapter,memory_tokens,memory_experts,adaln_memory,hybrid_interface,lora_memory\}_v1/report.json`	Fig. 8

The checked-in raw snapshot is `generated_results/raw_ogbench_snapshot.json`. It records the source report hash and embeds the aggregate result used by the manuscript. `generated_results/architecture_example_receipt.json` binds the selected environment, episode, timestamps, change scores, raw cache hash, and individual frame hashes.

B Raw Protocol Contract

Allowed inputs. The feature stage opens only 'frames' and 'actions' from each cache. Array indices are timestamps. The model does not consume reward, goal, simulator state, or any event annotation.

Disallowed inputs and procedures. The runtime source audit rejects calls named `draw_cue`, `inject_cue_sequence`, `inject_cue_sequence_mode`, or `_saliency_map`. Cached cue-label and cue-position arrays are listed as ignored. No fixed event interval, color rule, saturation rule, manually selected key frame, or semantic event label is used by feature extraction, host training, CEM training, threshold selection, or test inference.

Contract metadata. Every receipt and result stores `cue_window=null`, `cue_window_used_by_model=false`, `cue_metadata_used_by_model=false`, and the AST audit result. All 18 cells pass. The architecture frames are unmodified cache arrays; only display resizing is applied by the plotting library.

C Data Split and Feature Host

Each environment contains 768 trajectories, 22 frames per trajectory, and 21 actions. A fixed split seed, 20,260,721, assigns 538 trajectories to train, 115 to validation, and 115 to test. Split-index SHA256 hashes are stored in each feature receipt. The same split is used across optimization seeds.

The frozen visual encoder is DINOv2 ViT-S/14. Its local weights have SHA256 `b938bf1bc15cd2ec0feacfe3a1bb553fe8ea9ca46a7e1d8d00217f29aef60cd9`. The feature is a 1x1 plus 2x2 spatial pyramid over `x_norm_patchtokens`. PCA is fit on train trajectories only and retains 96 components. The finite host window contains two latent frames and aligned actions. The residual MLP predicts the next latent and is trained only on train trajectories. CEM training starts after the best validation host is restored and all host parameters are frozen.

D Raw CEM Training Details

Proposal and grouping. For each frame, the frozen host supplies one-step latent MSE and DINO features supply temporal semantic change. The 0.78 train quantile sets both thresholds. Adjacent

Table 4: Raw host quality. MSE is held-out one-step latent error. Host/persistence below one means the action-conditioned predictor improves on copying the latest latent. "Reliable" reports passed cells.

environment	n	host MSE	host/persistence	reliable
antmaze-large	1	0.2082	0.873	1/1
cube-double	1	0.3741	0.862	1/1
cube-single	3	0.3806	0.856	3/3
cube-triple	3	0.3519	0.872	3/3
humanoidmaze-large	1	0.1627	0.883	1/1
pointmaze-giant	1	0.2687	0.848	1/1
pointmaze-large	3	0.2761	0.866	3/3
pointmaze-teleport	3	0.3365	0.858	3/3
puzzle-3x3	1	0.5694	0.789	1/1
scene	1	0.4544	0.828	1/1

crossings form one group. If a trajectory has no crossing, its largest combined train-calibrated score supplies one provisional group; this fallback is automatic and label-free.

Semantic keys and versions. A train-only MiniBatchKMeans model clusters event vectors into at most 16 semantic keys. A provisional event remains separate from the active version with the same key. At its verification time, the new version is promoted only if its predicted score exceeds both the validation-selected threshold and the active score plus a hysteresis margin. A failed replacement leaves the active version unchanged.

Causal-effect target. For candidate group G, the calibration target is

$$y_G = \frac{\mathcal{L}_{\text{no memory}} - \mathcal{L}_{\text{singleton } G}}{\mathcal{L}_{\text{no memory}} + \epsilon}.$$

Deleting G from the singleton store returns the no-memory condition, so this is the normalized loss increase caused by group deletion. The estimator uses Smooth-L1 regression and within-query pairwise ranking. This target does not use a semantic event label.

Validation selection. The validation sweep covers promotion-score quantiles, two hysteresis margins, budgets 1/2/4, verification delays 1/3, and top-k values 1/2 where legal. The selected point minimizes future-latent MSE with a negligible retrieval tie-break cost. Test outcomes do not enter selection. Test-only budget curves add budget 8, and delay curves evaluate delays 1/2/3/4/6.

E Full Raw Controls and Telemetry

Table 5: Cell-level mean four-step test MSE over 18 cells. Every condition uses the same frozen host, trajectories, prediction target, and memory adapter.

condition	future-latent MSE
CEM	0.48383
no memory	0.48725
reset memory	0.48725
shuffled-episode memory	0.48939
matched-norm random memory	0.48952
recent-only state	0.48320

The mean relative CEM improvement over no memory is 0.719% with a normal cell-level 95% interval $[0.621, 0.818]\%$. The corresponding horizon reductions are 0.227%, 0.487%, 0.817%, and 1.086% for horizons one through four. All 10 environment means are positive. CEM is better than recent-only in 3/18 cells.

Mean proposals per query are 1.923. Delayed verification promotes 0.586 and rejects 0.414 of eligible groups. Mean active occupancy is 0.901 and retrieval rate is 0.803. The per-cell JSON additionally records superseded, fallback, evicted, and stale-selection counts.

F Causal Deletion and Action Use

For every test query with at least two candidate groups, the deletion test compares the predicted-high singleton group with a randomly selected candidate from the same query. Mean loss increases are 0.004253 and 0.004147. The high-minus-random mean is 0.000106 with interval $[-0.000056, 0.000269]$ and 12/18 cell wins. CE-score Spearman correlation is 0.037 on average, positive in 13/18 cells. These values support only a weak ranking claim.

The bounded retrieve-then-verify audit accepts 64.6% of routed candidates when the measured singleton group effect is positive. Its mean post-hoc verified-memory MSE is 0.48137. This audit is computed only after observing the future target and is not used by the primary prediction.

Action-sequence identification uses 256 test queries per cell where available. The observed four-action sequence is placed among seven sequences sampled from other episodes. Memory retrieval is selected with candidate actions masked, then each action sequence is rolled out. The score is latent error to the observed future. Mean top-1 accuracy is 0.3066 with memory and 0.3030 without memory, with 8/18 cell wins. This is an offline identification test, not model-predictive control.

G Official DINO-WM Validation

The separate official protocol uses checkpoint `outputs/dinowm_wall_audit_v1/checkpoint/model_latest.pth`, epoch 65, SHA256 `8441971becdae934fe08de5b163398390a32f6fe1fb0a8df113290e2468e142b`. The source revision is `0a9492fa12044b852ae9e001cc74604b79c8bb0c`. The checkpoint, action/proprioception encoders, and DINO features remain fixed.

Table 6: Released DINO-WM Wall factorial, means over seeds 0–2. Full/control is balanced accuracy; loss is memory/reset.

condition	overwrite	full/control	loss	high/random deletion
Immediate replacement	.219	.400/.243	1.311/1.397	.085/.000
Hysteresis only	.751	.774/.244	1.160/1.397	.237/.000
Versions, no verification	.551	.635/.236	1.223/1.397	.174/.000
CEM	.751	.758/.254	1.165/1.397	.296/.000

This protocol relocates cached genuine encoded events in latent time rather than rendering a raw environment stream. It validates the released host and the event-versioning mechanism, but it is not OGBench breadth, native planning, or evidence about unmodified event discovery.

H LeWM Memory Interface Details

All LeWM studies assert the same encoder/predictor digest before and after training: `5589632959b98370ad96001523025bc265686e82b87376d327da18cbd555f879`. Trainable adapters are separate from the fixed host.

Table 7: LeWM six-way PushT interface results, three-seed means. Intermediate columns report post-hoc task information. Strict ratio is future-latent loss relative to the unmodified host when objectives are comparable.

interface	memory	intermediate	host	strict ratio	result
Dense residual	.919	.169	.156	not comparable	interface loss
Memory experts	.921	.698	.370	36.79	prediction loss
Semantic bottleneck	.838	.760	.788	not comparable	diagnostic only
AdaLN	.961	.854	.813	31.08	prediction loss
Memory tokens	.984	.983	.819	13.59	prediction loss
Strict hybrid	.967	.840	.349	1.034	task information lost
LoRA	.742	.760	.185	.993	task information lost

The semantic bottleneck preserves the pairwise structure of counterfactual latent targets (distance Pearson 0.992), but its original cue-conditioned objective is not comparable to strict next-latent loss. Memory tokens and AdaLN make task information available but substantially worsen prediction. The strict hybrid and LoRA preserve prediction but do not make the binding available at the host output. No row satisfies both requirements.

I Decision-Log Schema

Each raw decision log records the environment, optimization seed, query timestamp, surprise stream, semantic-change stream, proposal groups, semantic key, event timestamp, predicted effect, router score, lifecycle transitions, promotion or rejection, supersession, eviction, fallback, retrieved event identifiers, and the selected store configuration. It also repeats the no-manual-event contract. Feature vectors are stored in checkpoints rather than expanded into JSON.

The architecture receipt uses a deterministic rule: among completed test logs, select the retrieved event with largest proposal score, then display the preceding frame, event frame, following frame, and query frame. For the current paper this selects HumanoidMaze-large, seed 0, episode 384, event timestamp 12, and query timestamp 16. The four raw frame hashes and full cache hash are in `generated_results/raw_architecture_receipt.json`.

J Reproduction

From the repository root:

```
.venv/bin/python -m pytest -q \
  scripts/test_run_cem_raw_ogbench.py

.venv/bin/python scripts/launch_cem_raw_ogbench.py \
  --smoke --gpus 1 2

.venv/bin/python scripts/launch_cem_raw_ogbench.py \
  --campaign --gpus 0 1 2

.venv/bin/python scripts/plot_cem_raw_ogbench.py
.venv/bin/python scripts/plot_paper_d.py

cd paper_d
/home/chrislin/.TinyTeX/bin/x86_64-linux/pdflatex \
  -interaction=nonstopmode -halt-on-error main.tex
/home/chrislin/.TinyTeX/bin/x86_64-linux/pdflatex \
  -interaction=nonstopmode -halt-on-error main.tex
```

The launcher accepts only physical GPUs 0, 1, and 2 and rejects GPU3. The completed launch receipt records no failed or running jobs. Generated figures, tables, and snapshots are intended to be included with the paper so a later PDF build does not require private output caches.

K Scope and Omitted Claims

We do not claim real-world camera generalization, robot deployment, native DINO-WM planning, executed raw-OGBench control, unbounded recall, cross-task transfer, or a well-calibrated causal-effect estimator. The breadth images are unmodified simulated environment renderings. Four environments have three optimization seeds and six have one. The recent-only comparison and weak deletion ordering are retained because they directly limit the main result.